PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Triple Negative Breast Cancer Prevalence in Africa: a Systematic Review and Meta-analysis
AUTHORS	Hercules, Shawn; Alnajar, Meena; Chen, Chen; Mladjenovic, Stefan; Shipeolu, Ajarat; Perkovic, Olga; Pond, G; Mbuagbaw, Lawrence; Blenman, Kim; Daniel, Juliet

VERSION 1 – REVIEW

REVIEWER	Ssentongo, Paddy Penn State Health Milton S Hershey Medical Center
REVIEW RETURNED	31-Aug-2021

GENERAL COMMENTS	Thank you for the invitation to review the manuscript entitled: Triple Negative Breast Cancer Prevalence in Africa: a Systematic Review and Meta-analysis. This meta-analysis pooled the frequency of TNBC across Africa. They found an overall prevalence of 27%. The prevalence was highest in West Africa (46%). Use of validated tool for assessing receptor expression, tool used for tumor biology assessment, and publication year were significant modifiers of the prevalence. Study quality was not associated with the prevalence. General comment: This is a well-conducted study and well-written manuscript tackling a critical topic of the burden of TNBC in Africa. Triple-negative breast cancer is very aggressive, and hence its burden in the African setting needs to be defined. The manuscript, however, can be improved by addressing the following comments below. Methods • Due to the considerable variation in the frequency of TNBC and the sample size, the back-transformation of the Freeman-Tukey double arcsine transformation may yield spurious results. I suggest the authors either use generalized linear mixed models (GLMM) or use logit transformation (see https://pubmed.ncbi.nlm.nih.gov/30945438/). At a minimum, the authors should compare their current results with the logit transformation and GLMM. • Please create a funnel plot to show the distribution of study effect sizes and to complement Figure S6. Results • Table 1 is difficult to read. I suggest reporting the mean age and SD and median age (IQR). Delete the min and max age row. For the

	grade, lymph node status, premenopausal frequency, and TBNC, report median and their corresponding IQR. • “Figure S5: Meta-regression shows association with the year of publication but not with age or use of a validated tool for receptor status” Please revise the sentence since the use of validated tool had a significant p-value. • Please show the results of the influence analysis. • Figure S4 is not in agreement with the p-value of Figure 5A. The confidence interval overlaps in Fig S4, indicative of a lack of significance. However, the p-value of 0.032 in Fig S5A. Please address the discrepancy.

REVIEWER	Pathak, Mona All India Institute of Medical Sciences, Biostatistics
REVIEW RETURNED	10-Nov-2021

GENERAL COMMENTS	The manuscript entitled “Triple Negative Breast Cancer Prevalence in Africa: a Systematic Review and Meta-analysis” estimated the prevalence of triple negative breast cancer (TNBC) in African female population which is really an important topic. The authors have found high heterogeneity in the estimated prevalence. Authors have provided estimates of TNBC prevalence for Central, East, North, Southern and West Africa and all those estimates also have high heterogeneity measured using I2 statistic (I2>87%). The subgroup analysis based on tool used to assess ER/PR status and HER2 status revealed that tools used for HER2 status may explain some degree of heterogeneity in TNBC prevalence. Authors have performed the meta-regression to identify the causes of heterogeneity in the pooled prevalence of TNBC such as age at diagnosis and validated or non-validated for receptor status. But these factors are not able to explain the heterogeneity in the estimated prevalence as these may be some genetic factors. However, this is also fact with systematic review and meta-analysis that authors can explore only those factors which has been explored by primary studies. Figure S5 and S6: In this manuscript authors are estimating TNBC prevalence not treatment effect so the y-axis of Figure S5 (A, B and C) and Figure S6 need to be labeled accordingly. Since there is high heterogeneity and publication bias, author should grade the confidence in their evidence using GRADE approach. I hope my queries will help in improving manuscript.

REVIEWER	Canelo-Aybar, Carlos Iberoamerican Cochrane Centre - IIB Sant Pau. CIBER de Epidemiología y Salud Pública (CIBERESP)
REVIEW RETURNED	17-Nov-2021

GENERAL COMMENTS	Relevant manuscript, please see some suggestions: 1) The PRISMA and MOOSE are reporting guidelines. I suggest that as a methodological guideline to conduct the review, you refer to other sources i.e. Cochrane Handbook. 2) The aim is to synthesize the studies on population prevalence. Given that please clarify why case-control studies and "other design beyond observational studies" were included?. Also for cohort

	studies, the data included is from the baseline (cross-sectional analysis) or from the longitudinal analysis? (period prevalence? which may require a separate analysis). ´Please clarify as case-control studies (unless nested) should not be included and may impact the validity of the results. 3) The statistical analysis is appropriate, but I would suggest using the logit transformation, and the Clopper-Pearson method to estimate the confidence interval for the proportion meta-analysis. Also, consider visual inspection to assess heterogeneity, as in some cases it might overestimate heterogeneity in proportion meta-analysis with narrow CIs.

REVIEWER	Marlink, Richard Rutgers
REVIEW RETURNED	19-Jan-2022

GENERAL COMMENTS	Well done.

VERSION 1 – AUTHOR RESPONSE

Additional Editorial Board Member Comments:

Reviewer #1: Dr. Paddy Ssentongo, Penn State Health Milton S Hershey Medical Center

Comments to the Author: Thank you for the invitation to review the manuscript entitled: Triple

Negative Breast Cancer Prevalence in Africa: a Systematic Review and Meta -analysis. This meta-analysis pooled the frequency of TNBC across Africa. They found an overall prevalence of 27%. The prevalence was highest in West Africa (46%). Use of validated tool for assessing receptor expression, tool used for tumor biology assessment, and publication year were significant modifiers of the prevalence. Study quality was not associated with the prevalence.

General comment: This is a well-conducted study and well-written manuscript tackling a critical topic of the burden of TNBC in Africa. Triple-negative breast cancer is very aggressive, and hence its burden in the African setting needs to be defined. The manuscript, however, can be improved by addressing the following comments below.

Methods

8. Due to the considerable variation in the frequency of TNBC and the sample size, the back-transformation of the Freeman-Tukey double arcsine transformation may yield spurious results. I suggest the authors either use generalized linear mixed models (GLMM) or use logit transformation (see https://pubmed.ncbi.nlm.nih.gov/30945438/). At a minimum, the

authors should compare their current results with the logit transformation and GLMM. ***Response:*** We have adjusted our statistical analyses which all use the logit transformation. We also conducted GLMM (data not shown) however as Schwarzer *et al.* (2019) states, individual study weights are not available for this method and for this purpose we opted for the logit transformation with inverse-variance method for pooling and Clopper-Pearson method for confidence intervals.

9. Please create a funnel plot to show the distribution of study effect sizes and to complement Figure S6.

Response: We have created a funnel plot to show the distribution of study effect sizes which is now depicted in Figure S8 on page 17 of the supplementary documents.

Results

10. Table 1 is difficult to read. I suggest reporting the mean age and SD and median age (IQR). Delete the min and max age row. For the grade, lymph node status, premenopausal frequency, and TBNC, report median and their corresponding IQR.

Response: We have amended Table 1 to report mean age (with SD), median age (with IQR), deleted min and max age row and added median for the remaining variables with IQR as suggested and this is reflected on page 11 of the manuscript in the Results section.

"Figure S5: Meta-regression shows association with the year of publication but not with age or use of a validated tool for receptor status" Please revise the sentence since the use of validated tool had a significant p-value.

Response: With our logit transformation, use of validated tool for receptor status did not yield statistical significance, hence we did not change this in our results.

12. Please show the results of the influence analysis.

Response: Influence analysis via Baujat plot has been reflected and depicted in Figure S6 on page 15 of the supplementary documents.

13. Figure S4 is not in agreement with the p-value of Figure 5A. The confidence interval overlaps in Fig S4, indicative of a lack of significance. However, the p-value of 0.032 in Fig S5A. Please address the discrepancy.

Response: We apologize for this discrepancy. With our logit transformation, we have updated all of our P values and the confidence interval in Fig. 5A was no longer significant when using logit.

Reviewer #2: Dr. Mona Pathak, All India Institute of Medical Sciences

Comments to the Author: The manuscript entitled "Triple Negative Breast Cancer Prevalence in Africa: a Systematic Review and Meta-analysis" estimated the prevalence of triple negative breast cancer (TNBC) in African female population which is really an important topic. The authors have found high heterogeneity in the estimated prevalence. Authors have provided estimates of TNBC prevalence for Central, East, North, Southern and West Africa and all those estimates also have high heterogeneity measured using I2 statistic (I2>87%). The subgroup analysis based on tool used to assess ER/PR status and HER2 status revealed that tools used for HER2 status may explain some degree of heterogeneity in TNBC prevalence. Authors have performed the meta-regression to identify the causes of heterogeneity in the pooled prevalence of TNBC such as age at diagnosis and validated or non-validated for receptor status. But these factors are not able to explain the heterogeneity in the estimated prevalence

as these may be some genetic factors. However, this is also fact with systematic review and meta-analysis that authors can explore only those factors which has been explored by primary studies.

14. Figure S5 and S6: In this manuscript authors are estimating TNBC prevalence not treatment effect so the y-axis of Figure S5 (A, B and C) and Figure S6 need to be labeled accordingly.

Response: We have revised and labeled these figures appropriately on the y-axis to reflect that effect estimates are being investigated on the y-axis for these meta-regressions.

15. Since there is high heterogeneity and publication bias, author should grade the confidence in their evidence using GRADE approach.

Response: We have updated our review to use the GRADE approach for confidence in our reported evidence as noted in our Methods section (on page 8), our Results section (on page 13) and our Discussion section (on page 16).

Reviewer #3: Dr. Carlos Canelo-Aybar, 3. Iberoamerican Cochrane Centre - IIB Sant Pau. CIBER de Epidemiologia y Salud Publica (CIBERESP), Universidad Peruana de Ciencias Aplicadas

Comments to the Author:

16. The PRISMA and MOOSE are reporting guidelines. I suggest that as a methodological guideline to conduct the review, you refer to other sources i.e. Cochrane Handbook.

Response: As suggested by Cochrane and in addition to PRISMA and MOOSE reporting guidelines, we have included the use of GRADE approach for reporting our confidence in the evidence reported in our review.

17. The aim is to synthesize the studies on population prevalence. Given that please clarify why case-control studies and "other design beyond observational studies" were included? Also for cohort studies, the data included is from the baseline (cross-sectional analysis) or from the longitudinal analysis? (period prevalence? which may require a separate analysis). Please clarify as case-control studies (unless nested) should not be included and may impact the validity of the results.

Response: Thank you for raising this important point. We planned to use any studies that would have “prevalence” data, irrespective of design. However, only one nested case-control study was included from which TNBC frequency could be estimated among the breast cancer cases reported. Controls were not included in any calculations for TNBC frequency.

18. The statistical analysis is appropriate, but I would suggest using the logit transformation, and the Clopper-Pearson method to estimate the confidence interval for the proportion

meta-analysis. Also, consider visual inspection to assess heterogeneity, as in some cases it might overestimate heterogeneity in proportion meta-analysis with narrow CIs.

Response: Logit transformation was used with the Clopper-Pearson method to estimate the confidence interval as suggested by this reviewer in all our meta-analyses shown. The authors thank you for this suggestion.

Reviewer# 4: Prof. Richard Marlink, Rutgers

Comments to the Author: Well done.

VERSION 2 – REVIEW

REVIEWER	Ssentongo, Paddy Penn State Health Milton S Hershey Medical Center
REVIEW RETURNED	16-Apr-2022

GENERAL COMMENTS	I commend the authors for their heroic efforts on the manuscript.